Lab 3-2 Network Address Translation

Learning Objectives

As a result of this lab section, you should achieve the following tasks:

- Translation of addresses between networks (NAT).
- Configuration of Easy IP.

Topology

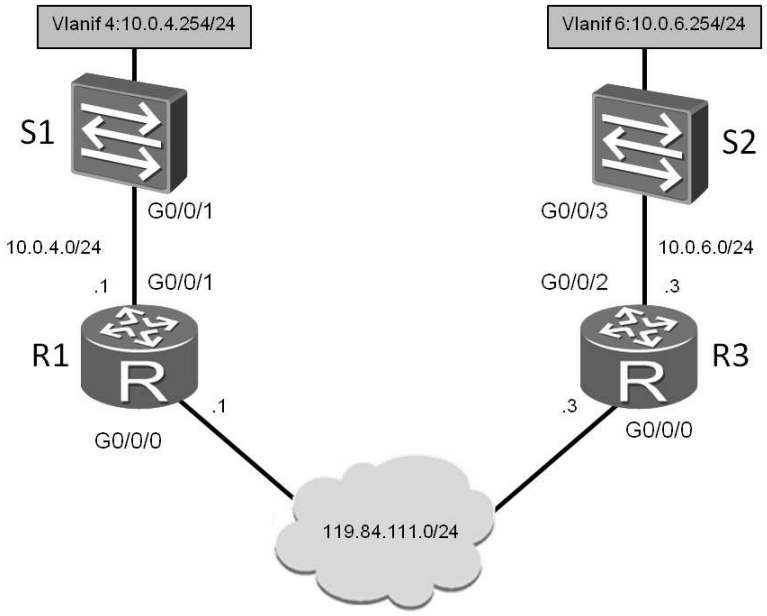


Figure 3.2 Network Address Translation Topology

Scenario

In order to conserve addressing the offices of the enterprise network have implemented private addressing internally. Users however require a means to be routed between these private networks and the public network domain. R1 and R3 represent edge routers of the enterprise branch offices ,the branch network need access to the public network. The administrator of the network is requested to configure dynamic NAT solutions on the in order to allow R1 to perform address translation. An

easyIP NAT solution is to be applied to R3.

Tasks

Step 1 Preparing the environment.

If you are starting this section with a non-configured device, begin here and then move to step 3. For those continuing from previous labs, begin at step 2.

```
[Huawei]sysname R1
[R1]inter GigabitEthernet0/0/1
[R1-GigabitEthernet0/0/1]ip address 10.0.4.1 24

[Huawei]sysname R3
[R3]interface GigabitEthernet0/0/2
[R3-GigabitEthernet0/0/2]ip address 10.0.6.3 24

[Huawei]sysname S1
[S1]vlan 4
[S1-vlan3]quit
[S1]interface vlanif 4
[S1-Vlanif4]ip address 10.0.4.254 24
[S1-Vlanif4]quit

[Huawei]sysname S2
[S2]vlan 6
[S2-vlan6]quit
[S2]interface vlanif 6
[S2-Vlanif6]ip address 10.0.6.254 24
[S2-Vlanif6]quit
```

Step 2 Clean up the previous configuration.

Re-establish the connection to S1 and S2 via Gigabit Ethernet 0/0/1 on R1 and Gigabit Ethernet 0/0/2 on R3. Remove OSPF from all routers.

```
[R1]interface GigabitEthernet 0/0/0
[R1-GigabitEthernet0/0/0]undo ip address
[R1]interface GigabitEthernet 0/0/1
[R1-GigabitEthernet0/0/1]undo shutdown
[R1]undo ospf 1
```

```
Warning: The OSPF process will be deleted. Continue? [Y/N]:y

[R2]undo ospf 1
Warning: The OSPF process will be deleted. Continue? [Y/N]:y

[R3-GigabitEthernet0/0/0]undo ip address
[R3]interface GigabitEthernet 0/0/2
[R3-GigabitEthernet0/0/2]undo shutdown
[R3]undo ospf 1
Warning: The OSPF process will be deleted. Continue? [Y/N]:y
```

Remove the static routes pointing to R2 on S1 and S2.

```
[S1]undo ip route-static 0.0.0.0 0.0.0.0

[S2]undo ip route-static 0.0.0.0 0.0.0.0
```

Step 3 **Implement VLAN configuration for S1 and S2.**

```
[S1]interface GigabitEthernet 0/0/1
[S1-GigabitEthernet0/0/1]port link-type trunk
[S1-GigabitEthernet0/0/1]port trunk pvid vlan 4
[S1-GigabitEthernet0/0/1]port trunk allow-pass vlan all
[S1-GigabitEthernet0/0/1]quit

[S2]interface GigabitEthernet 0/0/3
[S2-GigabitEthernet0/0/3]port link-type trunk
[S2-GigabitEthernet0/0/3]port trunk pvid vlan 6
[S2-GigabitEthernet0/0/3]port trunk allow-pass vlan all

[R1]interface GigabitEthernet0/0/0
[R1-GigabitEthernet0/0/0]ip address 119.84.111.1 24

[R3]interface GigabitEthernet0/0/0
[R3-GigabitEthernet0/0/0]ip address 119.84.111.3 24
```

Verify that R1 is able to reach both S1 and R3.

```
<R1>ping 10.0.4.254
  PING 10.0.4.254: 56  data bytes, press CTRL_C to break
    Reply from 10.0.4.254: bytes=56 Sequence=1 ttl=255 time=23 ms
    Reply from 10.0.4.254: bytes=56 Sequence=2 ttl=254 time=1 ms
    Reply from 10.0.4.254: bytes=56 Sequence=3 ttl=254 time=1 ms
    Reply from 10.0.4.254: bytes=56 Sequence=4 ttl=254 time=10 ms
    Reply from 10.0.4.254: bytes=56 Sequence=5 ttl=254 time=1 ms
```

```
  --- 10.0.4.254 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
     round-trip min/avg/max = 1/7/23 ms

<R1>ping 119.84.111.3
  PING 119.84.111.3: 56  data bytes, press CTRL_C to break
    Reply from 119.84.111.3: bytes=56 Sequence=1 ttl=255 time=1 ms
    Reply from 119.84.111.3: bytes=56 Sequence=2 ttl=255 time=10 ms
    Reply from 119.84.111.3: bytes=56 Sequence=3 ttl=255 time=1 ms
    Reply from 119.84.111.3: bytes=56 Sequence=4 ttl=255 time=1 ms
    Reply from 119.84.111.3: bytes=56 Sequence=5 ttl=255 time=10 ms

  --- 119.84.111.3 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
     round-trip min/avg/max = 1/4/10 ms
```

Step 4 **Configure Access Control Lists for R1 and R3.**

Configure an advanced ACL on R1 and select the data flow with the source of S1, the destination of R3, and destined for the telnet service port.

```
[R1]acl 3000
[R1-acl-adv-3000]rule 5 permit tcp source 10.0.4.254 0.0.0.0 destination
119.84.111.3 0.0.0.0 destination-port eq 23
[R1-acl-adv-3000]rule 10 permit ip source 10.0.4.0 0.0.0.255 destination any
[R1-acl-adv-3000]rule 15 deny ip
```

Configure a basic ACL on R3 and select the data flow whose source IP address is 10.0.6.0/24.

```
[R3]acl 2000
[R3-acl-basic-2000]rule permit source 10.0.6.0 0.0.0.255
```

Step 5 **Configure Dynamic NAT.**

Configure static route on S1 and S2,the nexthop as the private network’ s gateway.

```
[S1]ip route-static 0.0.0.0 0.0.0.0 10.0.4.1
[S2]ip route-static 0.0.0.0 0.0.0.0 10.0.6.3
```

Configure dynamic NAT on the GigabitEthernet0/0/0 interface of R1.

```
[R1]nat address-group 1 119.84.111.240 119.84.111.243
[R1]interface GigabitEthernet 0/0/0
[R1-GigabitEthernet0/0/0]nat outbound 3000 address-group 1
```

Configure R3 as the telnet server.

```
[R3]user-interface vty 0 4
[R3-ui-vty0-4]authentication-mode password
[R3-ui-vty0-4]set authentication password cipher huawei
[R3-ui-vty0-4]quit
```

Verify the address group has been configured correctly

```
<R1>display nat address-group
     NAT Address-Group Information:
     --------------------------------------
     Index   Start-address      End-address
     --------------------------------------
     1      119.84.111.240   119.84.111.243
     --------------------------------------
     Total : 1
```

Test connectivity to the gateway of the remote peer from the internal network.

```
<S1>ping 119.84.111.3
  PING 119.84.111.3: 56  data bytes, press CTRL_C to break
    Request time out
    Reply from 119.84.111.3: bytes=56 Sequence=2 ttl=254 time=1 ms
    Reply from 119.84.111.3: bytes=56 Sequence=3 ttl=254 time=1 ms
    Reply from 119.84.111.3: bytes=56 Sequence=4 ttl=254 time=1 ms
    Reply from 119.84.111.3: bytes=56 Sequence=5 ttl=254 time=1 ms

  --- 119.84.111.3 ping statistics ---
    5 packet(s) transmitted
    4 packet(s) received
    20.00% packet loss
    round-trip min/avg/max = 1/1/1 ms
```

Establish a telnet connection to the public address of the remote peer.

```
<S1>telnet 119.84.111.3
Trying 119.84.111.3 ...
Press CTRL+K to abort
Connected to 119.84.111.3 ...

Login authentication
```

```
Password:
<R3>
```

Do not exit the telnet session, instead open a second session window to R1 and view the results of the ACL and NAT session translation.

```
<R1>display acl 3000
Advanced ACL 3000, 2 rules
Acl's step is 5
rule 5 permit tcp source 10.0.4.254 0 destination 119.84.111.3 0 destination-port
eq telnet (1 matches)
rule 10 permit ip source 10.0.4.0 0.0.0.255 (1 matches)
rule 15 deny ip

<R1>display nat session all
  NAT Session Table Information:

    Protocol               : ICMP(1)
    SrcAddr   Vpn          : 10.0.4.254
    DestAddr  Vpn          : 119.84.111.3
    Type Code IcmpId       : 8   0   44003
    NAT-Info
      New SrcAddr          : 119.84.111.242
      New DestAddr         : ----
      New IcmpId           : 10247

    Protocol               : TCP(6)
    SrcAddr Port Vpn       : 10.0.4.254     49646
    DestAddr Port Vpn      : 119.84.111.3    23
    NAT-Info
      New SrcAddr          : 119.84.111.242
      New SrcPort          : 10249
      New DestAddr         : ----
      New DestPort         : ----

  Total : 2
```

The ICMP session has a lifetime of only 20 seconds and therefore may not appear to be present when displaying the NAT session results. The following command can be used in this case to extend the period over which the ICMP results are maintained:

```
[R1]firewall-nat session icmp aging-time 300
```

Configure easyIP on the Gigabit Ethernet 0/0/0 interface of R3, associating the easyIP configuration with ACL 2000 that had been configured earlier.

```
[R3-GigabitEthernet0/0/0]nat outbound 2000
```

Test the connectivity from S2 to R1 via R3.

```
<S2>ping 119.84.111.1
  PING 119.84.111.1: 56  data bytes, press CTRL_C to break
    Reply from 119.84.111.1: bytes=56 Sequence=1 ttl=254 time=1 ms
    Reply from 119.84.111.1: bytes=56 Sequence=2 ttl=254 time=1 ms
    Reply from 119.84.111.1: bytes=56 Sequence=3 ttl=254 time=1 ms
    Reply from 119.84.111.1: bytes=56 Sequence=4 ttl=254 time=1 ms
    Reply from 119.84.111.1: bytes=56 Sequence=5 ttl=254 time=1 ms

  --- 119.84.111.1 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
     round-trip min/avg/max = 1/1/1 ms

<R3>display acl 2000
Basic ACL 2000, 1 rule
Acl's step is 5
 rule 5 permit source 10.0.6.0 0.0.0.255 (1 matches)

<R3>display nat outbound acl 2000
 NAT Outbound Information:
--------------------------------------------------------------------------
Interface               Acl    Address-group/IP/Interface     Type
--------------------------------------------------------------------------
GigabitEthernet0/0/0    2000                   119.84.111.3    easyip
--------------------------------------------------------------------------
  Total : 1
```

Final Configuration

```
<R1>display current-configuration
[V200R003C00SPC200]
#
```

```
 sysname R1
#
firewall-nat session icmp aging-time 300
#
acl number 3000
 rule 5 permit tcp source 10.0.4.254 0 destination 119.84.111.3 0 destination-port
eq telnet
 rule 10 permit ip source 10.0.4.0 0.0.0.255
 rule 15 deny ip
#
 nat address-group 1 119.84.111.240 119.84.111.243
#
interface GigabitEthernet0/0/0
 ip address 119.84.111.1 255.255.255.0
 nat outbound 3000 address-group 1
#
interface GigabitEthernet0/0/1
 ip address 10.0.4.1 255.255.255.0
#
user-interface con 0
 authentication-mode password
 set authentication password
cipher %$%$dD#}P<HzJ;Xs%X>hOkm!,.+Iq61QK`K6tI}cc-;k_o`C.+L,%$%$
user-interface vty 0 4
#
return

<R3>display current-configuration
[V200R003C00SPC200]
#
 sysname R3
#
acl number 2000
 rule 5 permit source 10.0.6.0 0.0.0.255
#
interface GigabitEthernet0/0/0
 ip address 119.84.111.3 255.255.255.0
 nat outbound 2000
#
interface GigabitEthernet0/0/2
 ip address 10.0.6.3 255.255.255.0
#
user-interface con 0
```

```
 authentication-mode password
 set authentication password
cipher %$%$W|$)M5D}v@bY^gK\;>QR,.*d;8Mp>|+EU,:~D~8b59~..*g,%$%$
user-interface vty 0 4
 authentication-mode password
 set authentication password
cipher %$%$7ml|,!ccE$SQ~CZ{GtaE%hO>v}~bVk18p5qq<:UPtI:9hOA%%$%$
#
return

<S1>display current-configuration
#
!Software Version V100R006C00SPC800
 sysname S1
#
 vlan batch 4
#
interface Vlanif4
 ip address 10.0.4.254 255.255.255.0
#
interface GigabitEthernet0/0/1
 port link-type trunk
 port trunk pvid vlan 4
 port trunk allow-pass vlan 2 to 4094
#
interface GigabitEthernet0/0/2
 port link-type trunk
 port trunk pvid vlan 4
 port trunk allow-pass vlan 2 to 4094
#
interface GigabitEthernet0/0/14
 shutdown
#
 ip route-static 0.0.0.0 0.0.0.0 10.0.4.1
#
user-interface con 0
user-interface vty 0 4
 set authentication password cipher N`C55QK<`=/Q=^Q`MAF4<1!!
#
return

<S2>display current-configuration
#
```

```
!Software Version V100R006C00SPC800
 sysname S2
#
 vlan batch 6
#
interface Vlanif6
 ip address 10.0.6.254 255.255.255.0
#
interface GigabitEthernet0/0/2
 port link-type trunk
 port trunk pvid vlan 6
 port trunk allow-pass vlan 2 to 4094
#
interface GigabitEthernet0/0/3
 port link-type trunk
 port trunk pvid vlan 6
 port trunk allow-pass vlan 2 to 4094
#
interface GigabitEthernet0/0/23
 shutdown
#
 ip route-static 0.0.0.0 0.0.0.0 10.0.6.3
#
user-interface con 0
user-interface vty 0 4
#
return
```